

homework 1.1-1.4

chl

2026 年 3 月 13 日

2.1.1.

设集合 $E \subset R^p$, 映射 $\varphi: E \rightarrow R^q$ 连续。证明对于 R^p 的任意闭集 $F \subset E$, 象集 $\varphi(F)$ 是一个 F_σ 型集 (从而是 Borel 集)。

证明. Consider

$$F = \bigcup_{n=1}^{\infty} F \cap \overline{B(0, n)}$$

Since F is a closed set and $\overline{B(0, n)}$ is a compact set, $F \cap \overline{B(0, n)}$ is a compact set. Since φ is continuous, $\varphi(F \cap \overline{B(0, n)})$ is a compact set. Therefore,

$$\varphi(F) = \bigcup_{n=1}^{\infty} \varphi(F \cap \overline{B(0, n)})$$

is a countable union of closed sets, i.e. a F_σ set. □

2.1.2.

对任意集合 $X \subset R^d$, 证明 $B(X) = \{X \cap A \mid A \in B(R^d)\}$ 。若进一步假设 $X \in B(R^d)$, 证明 $B(X) = \{E \in B(R^d) \mid E \subset X\}$ 。

证明. Let $\mathcal{A} = \{X \cap A \mid A \in B(\mathcal{R}^d)\}$. First we prove $\mathcal{A}$ is a σ -algebra containing all relatively open subsets of X , i.e. $B(X) \subset \mathcal{A}$.

1. Since $\mathcal{R}^d \in B(\mathcal{R}^d)$, $X = X \cap \mathcal{R}^d \in \mathcal{A}$.
2. $\forall$ relative open set A , $\exists$ open set $G \subset \mathcal{R}^d$, then $G \in B(\mathcal{R}^d)$, $A = G \cap X \in \mathcal{A}$.
3. $\forall A = X \cap B \in \mathcal{A}$, $B \in B(\mathcal{R}^d)$, $B^c \in B(\mathcal{R}^d)$, $X \setminus A = X \cap (X^c \cup B^c) = X \cap B^c \in \mathcal{A}$.
4. $\forall B_n = X \cap A_n \in \mathcal{A}$, $A_n \in B(\mathcal{R}^d)$, then $\bigcup_n A_n \in B(\mathcal{R}^d)$, $\bigcup_n B_n = X \cap \bigcup_n A_n \in \mathcal{A}$.

We next prove $B(X) \supset \mathcal{A}$, i.e. $\forall A \in B(\mathcal{R}^d)$ $X \cap A \in B(X)$. This equals to proving

$$\mathcal{C} \triangleq \{A \subset \mathcal{R}^d : X \cap A \in B(X)\} \supset B(\mathcal{R}^d)$$

We only need to prove that $\mathcal{C}$ is a Borel σ -algebra.

1. $\mathcal{R}^d \cap X = X \in B(X) \implies \mathcal{R}^d \in \mathcal{C}$.
2. $\forall$ open set $A \subset \mathcal{R}^d$, $X \cap A$ is a relative open set of X , $X \cap A \in \mathcal{C}$.
3. $\forall A \in \mathcal{C}$, $X \cap A \in B(X)$, $X \setminus (X \cap A) = X \cap A^c \in B(X)$, $A^c \in \mathcal{C}$.
4. $\forall A_n \in \mathcal{C}$, $X \cap A_n \in B(X)$, $X \cap \bigcup_n A_n = \bigcup_n X \cap A_n \in B(X)$, $\bigcup_n A_n \in \mathcal{C}$.

Then $\mathcal{C}$ is a σ -algebra and contains all the open sets of $\mathcal{R}^d$, then $\mathcal{C}$ is a Borel σ -algebra $\implies \mathcal{C} \supset B(\mathcal{R}^d)$.

Therefore, $B(X) = \mathcal{A}$.

Assume that $X \in B(\mathcal{R}^d)$, it is trivial that $B(X) = \mathcal{A} \subset \{E \in B(\mathcal{R}^d) \mid E \subset X\}$. We next prove reverse inclusion.

$\forall X \supset E \in B(\mathcal{R}^d)$, $E = E \cap X \in \mathcal{A} = B(X) \implies B(X) = \mathcal{A} \supset \{E \in B(\mathcal{R}^d) \mid E \subset X\}$. Therefore, $B(X) = \{E \in B(\mathcal{R}^d) \mid E \subset X\}$. $\square$

2.2.1 (测度的等价定义) .

设 (X, M) 为一可测空间, 设非负函数 $\mu : M \rightarrow [0, \infty]$ 满足测度定义中的 (ii) 即满足可数可加性。证明

$$\mu \text{ 是 } X \text{ 上的测度} \iff \exists A \in M \text{ 使得 } \mu(A) < \infty.$$

证明. This is equivalent to

$$\mu(\emptyset) = 0 \iff \exists A \in M \text{ s.t. } \mu(A) < \infty$$

$\mu(\emptyset) = 0 < \infty$. Assume that $\exists A \in M$ s.t. $\mu(A) < \infty$. Let $E_1 = A, E_n = \emptyset, n \geq 2$, then $\forall i \neq j, E_i \cap E_j = \emptyset$,

$$\mu(A) = \mu\left(\bigcup_n E_n\right) = \mu(A) + \sum_{n=2}^{\infty} \mu(\emptyset) < \infty \implies \mu(\emptyset) = 0$$

□

2.2.2 (测度空间的遗传) .

设 (X, M, μ) 为一测度空间, 设

$$X_0 \in M, \quad M_0 = \{E \in M \mid E \subset X_0\}, \quad \mu_0 = \mu|_{M_0}.$$

证明 (X_0, M_0, μ_0) 也是一个测度空间. 此外若 (X, M, μ) 是完备的, 则 (X_0, M_0, μ_0) 也是完备的。

【注】 由这一性质我们看到, 在 X 的某个可测子集 $X_0 \in M$ 上考虑可测性或可积性问题, 与在“整个测度空间” (X, M, μ) 上考虑可测性或可积性问题, 二者在概念上是等价的, 因为 (X, M, μ) 的子空间 (X_0, M_0, μ_0) 也是一个“整个测度空间”。

证明. Since μ_0 trivially satisfies the definition of measure, we only need to prove (X_0, M_0) is a measurable space, i.e. M is a σ -field of X_0 :

1. $X_0 \in M, X_0 \subset X_0 \implies X_0 \in M_0$
2. $\forall E \in M_0, E \subset X_0$, then $X_0 \setminus E = X_0 \cap E^c = (X_0^c \cup E)^c \in M, X_0 \setminus E \subset X_0$, then $X_0 \setminus E \in M_0$.
3. $\forall E_n \in M_0, E_n \subset X_0$, then $\bigcup_n E_n \subset X_0, \bigcup_n E_n \in M$, then $\bigcup_n E_n \in M_0$.

We assume that (X, M, μ) is complete. We next show (X_0, M_0, μ_0) is complete:

$\forall E \in M_0 \subset M$ satisfying $\mu(E) = \mu_0(E) = 0$, then $\forall A \subset E, \forall T \subset X_0$. Since (X, M, μ) is complete, $A \in M$. Considering $A \subset E \subset X_0, A \in M_0 \implies (X_0, M_0, \mu_0)$ is complete. Also,

$$0 \leq \mu_0(A) \leq \mu_0(E) = \mu(E) = 0 \implies \mu_0(A) = 0$$

□

2.2.3.

设 (X, M, μ) 是一个 σ -有限的测度空间且 $\mu(X) = \infty$ 。证明存在一系列 M -可测集 E_n 满足 $\mu(E_n) < \infty \forall n \in \mathbb{N}$ 且 $\mu(E_n) \rightarrow \infty (n \rightarrow \infty)$ 。

以下 2.2.4-2.2.9 题中的可测集、 σ -代数 M 和测度 μ 均取自测度空间 (X, M, μ) 。

证明. Since the measurable space is σ -finite, $\exists A_n \in M, \bigcup_n A_n = X, \mu(A_n) < \infty$. Let $E_n = \bigcup_{k=1}^n A_k$, then $\forall n \in \mathbb{N}, \mu(E_n) \leq \sum_{k=1}^n \mu(A_k) < \infty$. Since $E_n \uparrow X, \mu(E_n) \rightarrow \mu(X) = \infty (n \rightarrow \infty)$. $\square$

2.2.4.

设 $A, B \in M$ 。下列等式中哪一个可能无意义? 何时有意义?

$$\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B), \quad \mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

证明. The first equality is always meaningful.

$$\mu(A \cup B) + \mu(A \cap B) = \mu(A + B \cap A^c) + \mu(B \cap A) = \mu(A) + \mu(B \cap A^c) + \mu(B \cap A) = \mu(A) + \mu(B)$$

The second equality is meaningful if and only if $\mu(A \cap B) < \infty$.

If $\mu(A \cap B) < \infty$, the equality trivial holds and is meaningful.

To make the second equality meaningful, we need either $\mu(A \cap B) < \infty$ or $\max\{\mu(A), \mu(B)\} < \infty \implies \mu(A \cap B) < \infty$. $\square$

2.2.5.

设 $E_1, E_2, \dots, E_n$ 为可测集。

(1) 设 $E = E_1 \cup E_2 \cup \dots \cup E_n$, 则存在 $k \in \{1, 2, \dots, n\}$ 使得 $\mu(E_k) \geq \frac{1}{n}\mu(E)$ 。

(2) 若 $\mu(E_1 \cup E_2 \cup \dots \cup E_n) > n$, 则存在 $k \in \{1, 2, \dots, n\}$ 使得 $\mu(E_k) > 1$ 。(鸽笼原理)

(1)

证明. If $\exists 1 \leq k \leq n$, $\mu(E_k) = \infty$, then $\mu(E) \geq \mu(E_k) = \infty$. The inequality holds.

If $\forall 1 \leq k \leq n$, $\mu(E_k) < \infty$, then $\exists 1 \leq N \leq n$, $\mu(E_N) = \max_{1 \leq k \leq n} \mu(E_k)$,

$$\mu(E) \leq \sum_{k=1}^n \mu(E_k) \leq n\mu(E_N) \implies \mu(E_N) \geq \frac{1}{n}\mu(E)$$

□

(2)

证明. If $\exists 1 \leq k \leq n$, $\mu(E_k) = \infty$, then $\mu(E_k) > 1$. The inequality holds.

If $\forall 1 \leq k \leq n$, $\mu(E_k) < \infty$, then by (1), $\mu(E_k) \geq \frac{1}{n}\mu(E) > 1$.

□

2.2.6.

设 $E, E_k \in M$ 且 $E_k \subset E$, $k = 1, 2, \dots, n$; 满足 $\mu(E) = 1$, $\sum_{k=1}^n \mu(E_k) > n - 1$ 。

证明

$$\mu\left(\bigcap_{k=1}^n E_k\right) > 0.$$

证明. Since $\forall k$, $\mu(E_k) \leq \mu(E) = 1 < \infty$,

$$\begin{aligned} 1 - \mu\left(\bigcap_{k=1}^n E_k\right) &= \mu(E \setminus \bigcap_{k=1}^n E_k) = \mu\left(\bigcup_{k=1}^n (E \setminus E_k)\right) \\ &\leq \sum_{k=1}^n \mu(E \setminus E_k) = \sum_{k=1}^n (\mu(E) - \mu(E_k)) \quad (E \supset E_k, \forall k) \\ &= \sum_{k=1}^n \mu(E) - \sum_{k=1}^n \mu(E_k) < n - (n - 1) = 1 \\ &\implies \mu\left(\bigcap_{k=1}^n E_k\right) > 0 \end{aligned}$$

□

2.2.7.

设 $E_k, E \in M$ 满足 $E_k \subset E$, $\mu(E_k) = \mu(E) < \infty$, $k = 1, 2, 3, \dots$ 。证明

$$\mu\left(E \setminus \bigcap_{k=1}^{\infty} E_k\right) = 0, \quad \mu\left(\bigcap_{k=1}^{\infty} E_k\right) = \mu(E).$$

证明. $\forall k$, since $E \supset E_k$, $\mu(E_k) = \mu(E) < \infty$, $k = 1, 2, 3, \dots$, computing

$$0 \leq \mu\left(E \setminus \bigcap_{k=1}^{\infty} E_k\right) = \mu\left(\bigcup_{k=1}^{\infty} (E \setminus E_k)\right) \leq \sum_{k=1}^{\infty} \mu(E \setminus E_k) = \sum_{k=1}^{\infty} (\mu(E) - \mu(E_k)) = 0$$

gives $\mu(E \setminus \bigcap_{k=1}^{\infty} E_k) = 0$, and

$$0 = \mu\left(E \setminus \bigcap_{k=1}^{\infty} E_k\right) = \mu(E) - \mu\left(\bigcap_{k=1}^{\infty} E_k\right) \implies \mu\left(\bigcap_{k=1}^{\infty} E_k\right) = \mu(E)$$

□

2.2.8.

设 $E, E_n \in M$, $n = 1, 2, 3, \dots$ 满足

$$0 < \mu(E) < \infty, \quad E_n \subset E, \quad n = 1, 2, 3, \dots; \quad \limsup_{n \rightarrow \infty} \mu(E_n) = \mu(E).$$

证明对任意 $\varepsilon > 0$, 存在子列 $\{E_{n_k}\}_{k=1}^{\infty}$ 使得

$$\mu\left(\bigcap_{k=1}^{\infty} E_{n_k}\right) > \mu(E) - \varepsilon.$$

证明. $\forall \varepsilon > 0$, $\forall k \in \mathbb{N}$, $\exists n_k \in \mathbb{N}$ s.t.

$$n_1 < n_2 < \dots \quad \mu(E_{n_k}) > \mu(E) - \frac{\varepsilon}{2^k}.$$

Since $E_{n_k} \subset E$, $\mu(E) < \infty$, then

$$\mu(E \setminus E_{n_k}) = \mu(E) - \mu(E_{n_k}) < \frac{\varepsilon}{2^k}, \quad k = 1, 2, 3, \dots$$

Therefore,

$$\begin{aligned}\mu\left(E \setminus \bigcap_{k=1}^{\infty} E_{n_k}\right) &= \mu\left(\bigcup_{k=1}^{\infty} (E \setminus E_{n_k})\right) \leq \sum_{k=1}^{\infty} \mu(E \setminus E_{n_k}) < \sum_{k=1}^{\infty} \frac{\epsilon}{2^k} = \epsilon. \\ \implies \mu\left(\bigcap_{k=1}^{\infty} E_{n_k}\right) &= \mu(E) - \mu\left(E \setminus \bigcap_{k=1}^{\infty} E_{n_k}\right) > \mu(E) - \epsilon.\end{aligned}$$

□

2.2.9 (Borel-Cantelli 引理) .

设 $E_k \in M$, $k = 1, 2, 3, \dots$. 令

$$E = \{x \in X \mid x \text{ 同时属于无限多个 } E_k\}.$$

则 E 是可测集。具体来说, 这集合 E 可以等价地写成

$$E = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k.$$

若进一步假设

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty,$$

则有 $\mu(E) = 0$ 。

[说明: “ x 同时属于无限多个 E_k ”等价于“存在与 x 有关的子列 $\{E_{k_n}\}_{n=1}^{\infty}$ 使得 $x \in \bigcap_{n=1}^{\infty} E_{k_n}$ ”。
注意: 根据子列的定义知 $1 \leq k_1 < k_2 < k_3 < \dots$.]

证明. $\forall x \in E$, $\exists \{E_{k_n}\}_{n=1}^{\infty}$ s.t. $x \in \bigcap_{n=1}^{\infty} E_{k_n}$. Then $\forall N \in \mathbb{N}$, $\exists n \in \mathbb{N}$, $k_n > N$,

$$x \in E_{k_n} \subset \bigcup_{k=N}^{\infty} E_k \implies x \in \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$$

Therefore, $E \subset \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$.

$\forall x \in \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$, $\forall n \in \mathbb{N}$, $x \in \bigcup_{k=n}^{\infty} E_k$, then $\exists k = k_n \geq n$, s.t. $x \in E_k = E_{k_n}$. Since $\lim_{n \rightarrow \infty} k_n = \infty$, we can get a subsequence from it as $\{k_{n_j}\}_{j=1}^{\infty}$ satisfying $\forall j' > j$, $k_{n_{j'}} > k_{n_j}$. Write $k_j = k_{n_j}$. Now we get the subset $\{E_{k_n}\}_{n=1}^{\infty}$ where $\lim_{n \rightarrow \infty} k_n = \infty$ and $x \in \bigcap_{n=1}^{\infty} E_{k_n}$. Therefore, $E \supset \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$.

Based on the above, $E = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$.

Let $A_n = \bigcup_{k=n}^{\infty} E_k$, then $A_n \supset A_{n+1}$, i.e. $A_n \downarrow E = \bigcap_{n=1}^{\infty} A_n$. Since $\sum_{k=1}^{\infty} \mu(E_k) < \infty$, we have $\lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} \mu(E_k) = 0$. Therefore,

$$0 \leq \mu(E) \leq \mu(A_n) \leq \sum_{k=n}^{\infty} \mu(E_k) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

This gives $\mu(E) = 0$. □